C1.3 What is the evidence for climate change, why is it occurring?

Teaching and learning narrative

Some electromagnetic radiation from the Sun passes through the atmosphere and is absorbed by the Earth warming it. The warm Earth emits infrared radiation which some gases, including carbon dioxide and methane, absorb and re-emit in all directions; this keeps the Earth warmer than it would otherwise be and is called the greenhouse effect. Without the greenhouse effect the Earth would be too cold to support life.

The proportion of greenhouse gases in the Earth's atmosphere has increased over the last 200 years as a result of human activities. There are correlations between changes in the composition of the atmosphere, consumption of fossil fuels and global temperatures over time. Although there are uncertainties in the data, most scientists now accept that recent climate change can be explained by increased greenhouse gas emissions.

Patterns in the data have been used to propose models to predict future climate changes. As more data is collected, the uncertainties in the data decrease, and our confidence in models and their predictions increases (IaS3).

Scientists aim to reduce emissions of greenhouse gases, for example by reducing fossil fuel use and removing gases from the atmosphere by carbon capture and reforestation. These actions need to be supported by public regulation. Even so, it is difficult to mitigate the effect of emissions due to the very large scales involved. Each new measure may have unforeseen impacts on the environment, making it difficult to make reasoned judgments about benefits and risks (IaS4).

Assessable learning outcomes

Learners will be required to:

1. describe the greenhouse effect in terms of the interaction of radiation with matter
2. evaluate the evidence for additional anthropogenic causes of climate change, including the correlation between change in atmospheric carbon dioxide concentration and the consumption of fossil fuels, and describe the uncertainties in the evidence base
3. describe the potential effects of increased levels of carbon dioxide and methane on the Earth's climate, including where crops can be grown, extreme weather patterns, melting of polar ice and flooding of low land
4. describe how the effects of increased levels of carbon dioxide and methane may be mitigated, including consideration of scale, risk and environmental implications
5. extract and interpret information from charts, graphs and tables
   M2c, M4a
6. use orders of magnitude to evaluate the significance of data
   M2h

Linked learning opportunities

Specification links:

- What is global warming and what is the evidence for it? (P1.3)

Practical work:

- Investigate climate change models – both physical models and computer models

Ideas about Science:

- Use ideas about correlation and cause, about models and the way science explanations are developed when discussing climate change (IaS3)
- Risks, costs and benefits of fuel use and its sustainability and effects on climate (IaS4)
- public regulation of targets for emissions and reasons why different decisions on issues related to climate change might be made in view of differences in personal, social, or economic context (IaS4)